**Information Theory and Its Application in Deep Learning: Final Solutions**

1. **Differential Entropy and Maximum Entropy** (*30 points*)
   Let $X$ be a continuous random variable. Recall $h(X) = -\int f(x) \log f(x)\, dx$ (nats).

   (a) Let $X \sim \mathrm{Exp}(\lambda)$, i.e. $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Compute $h(X)$. (*10 points*)

   (b) Let $Y = cX$ for a constant $c > 0$. Find $h(Y)$ in two ways: (i) from the scaling rule, and (ii) by identifying the distribution of $Y$. (*10 points*)

   (c) Show that among *all* densities supported on $(0, \infty)$ with mean $\mathbb{E}[X] = \mu$, the exponential density $f(x) = \frac{1}{\mu} e^{-x/\mu}$ maximizes $h$. (*10 points*)

**Solution: Differential Entropy and Maximum Entropy.**

(a) Take the logarithm of the density first: $\log f(x) = \log \lambda - \lambda x$. Substituting into the definition and splitting the integral,

$$\begin{aligned} h(X) &= -\int_0^\infty \lambda e^{-\lambda x} \big( \log \lambda - \lambda x \big)\, dx \\ &= -\log \lambda \underbrace{\int_0^\infty \lambda e^{-\lambda x}\, dx}_{=1} + \lambda \underbrace{\int_0^\infty x\, \lambda e^{-\lambda x}\, dx}_{=\mathbb{E}[X]} . \end{aligned}$$

The first integral is 1 because $f$ is a density; the second is $\mathbb{E}[X] = \frac{1}{\lambda}$. Hence

$$h(X) = -\log \lambda + \lambda \cdot \tfrac{1}{\lambda} = 1 - \log \lambda.$$

(b) **(i) Scaling rule.** For $Y = cX$ with $c > 0$ the density transforms as $f_Y(y) = \frac{1}{c} f_X(y/c)$, so

$$h(Y) = -\int f_Y(y) \log f_Y(y)\, dy = -\int \tfrac{1}{c} f_X(y/c) \big[ \log f_X(y/c) - \log c \big]\, dy.$$

Substituting $x = y/c$ (so $dy = c\, dx$) gives $h(Y) = -\int f_X(x) \big[ \log f_X(x) - \log c \big]\, dx = h(X) + \log c$. Therefore $h(Y) = 1 - \log \lambda + \log c$.
**(ii) Identify the law.** If $X \sim \mathrm{Exp}(\lambda)$ then $Y = cX$ has $\Pr(Y \leq y) = \Pr(X \leq y/c) = 1 - e^{-\lambda y/c}$, i.e. $Y \sim \mathrm{Exp}(\lambda/c)$. By part (a) with rate $\lambda/c$, $h(Y) = 1 - \log(\lambda/c) = 1 - \log \lambda + \log c$. The two answers agree.

(c) Let $g$ be any density on $(0, \infty)$ with $\int_0^\infty x\, g(x)\, dx = \mu$, and let $f(x) = \frac{1}{\mu} e^{-x/\mu}$ be the exponential with the same mean. Start from non-negativity of KL and expand the integrand:

$$0 \leq D(g \| f) = \int g \log \frac{g}{f} = \int g \log g - \int g \log f = -h(g) - \int g(x) \log f(x)\, dx.$$

The key point is that $\log f$ is *linear* in $x$: $\log f(x) = -\log \mu - x/\mu$, so its $g$-average depends on $g$ only through the mean,

$$\int g(x) \log f(x)\, dx = -\log \mu - \frac{1}{\mu} \int_0^\infty x\, g(x)\, dx = -\log \mu - 1,$$

which is the same value it takes under $f$ itself. Substituting back, $0 \le -h(g) + \log \mu + 1$, i.e.

$$h(g) \le 1 + \log \mu.$$

The bound is attained by $f$, since $h(f) = 1 - \log(1/\mu) = 1 + \log \mu$ (part (a) with $\lambda = 1/\mu$). Hence the exponential maximizes $h$ over this class, with maximum entropy $1 + \log \mu$, and equality holds iff $D(g\|f) = 0$, i.e. $g = f$.

2. **Rate–Distortion of an Independent Gaussian Pair** (*35 points*)
Consider a vector source $\mathbf{X} = (X_1, X_2)$ with *independent* components $X_i \sim \mathcal{N}(0, \sigma_i^2)$, and squared-error distortion $d(\mathbf{x}, \hat{\mathbf{x}}) = \sum_{i=1}^{2}(x_i - \hat{x}_i)^2$. Work with the information rate–distortion function

$$R(D) = \min_{p(\hat{\mathbf{x}}|\mathbf{x}):\, \mathbb{E}\|\mathbf{X}-\hat{\mathbf{X}}\|^2 \le D} I(\mathbf{X}; \hat{\mathbf{X}}),$$

and write $R_i$ for the single-component function $R_i(D_i) = \left[\frac{1}{2}\log\frac{\sigma_i^2}{D_i}\right]^+$, i.e. $\frac{1}{2}\log(\sigma_i^2/D_i)$ for $0 < D_i \le \sigma_i^2$, 0 for $D_i \ge \sigma_i^2$, and $R_i(0) = +\infty$. Rates are in bits, $\log = \log_2$. The goal is to prove

$$R(D) = \min_{\substack{D_1, D_2 \ge 0 \\ D_1 + D_2 \le D}} \big(R_1(D_1) + R_2(D_2)\big).$$

(a) **Decoupling lemma.** (*10 points*) For *any* joint distribution of $(\mathbf{X}, \hat{\mathbf{X}})$, show

$$I(\mathbf{X}; \hat{\mathbf{X}}) \;\ge\; I(X_1; \hat{X}_1) + I(X_2; \hat{X}_2).$$

*Hint.* Write $I = h(\mathbf{X}) - h(\mathbf{X} \mid \hat{\mathbf{X}})$; use independence on the first term, the chain rule on the second, and "conditioning reduces entropy."

(b) **Converse.** (*10 points*) Show that

$$R(D) \;\ge\; \min_{D_1 + D_2 \le D} \big(R_1(D_1) + R_2(D_2)\big).$$

*Hint.* Take any test channel feasible for $R(D)$; set $D_i = \mathbb{E}[(X_i - \hat{X}_i)^2]$ so $D_1 + D_2 \le D$. Apply part (a), then bound each $I(X_i; \hat{X}_i)$ below using the definition of the single-letter $R_i$.

(c) **Achievability.** (*15 points*) Show the matching upper bound

$$R(D) \;\le\; \min_{D_1 + D_2 \le D} \big(R_1(D_1) + R_2(D_2)\big),$$

so that equality holds.
*Hint.* Fix any split $(D_1, D_2)$ with $D_1 + D_2 \le D$ and build a *product* test channel $p(\hat{\mathbf{x}} \mid \mathbf{x}) = \prod_i p_i^\star(\hat{x}_i \mid x_i)$ from the per-component optimizers; compute its mutual information and its distortion.

**Solution: Rate–Distortion of an Independent Gaussian Pair.**

(a) **(Decoupling lemma.)** Write the mutual information in differential-entropy form, $I(\mathbf{X};\hat{\mathbf{X}}) = h(\mathbf{X}) - h(\mathbf{X} \mid \hat{\mathbf{X}})$, and treat the two terms separately.

*First term.* Since $X_1 \perp X_2$, the joint density factorizes and the entropy is additive: $h(\mathbf{X}) = h(X_1) + h(X_2)$.

*Second term.* By the chain rule for differential entropy,

$$h(\mathbf{X} \mid \hat{\mathbf{X}}) = h(X_1 \mid \hat{\mathbf{X}}) + h(X_2 \mid X_1, \hat{\mathbf{X}}).$$

Conditioning on more variables can only reduce differential entropy. Dropping variables from each conditioning set (keep only $\hat{X}_1$ in the first term, only $\hat{X}_2$ in the second) gives the upper bound

$$h(X_1 \mid \hat{\mathbf{X}}) \le h(X_1 \mid \hat{X}_1), \qquad h(X_2 \mid X_1, \hat{\mathbf{X}}) \le h(X_2 \mid \hat{X}_2),$$

because $\hat{\mathbf{X}} = (\hat{X}_1, \hat{X}_2)$ contains $\hat{X}_1$, and $(X_1, \hat{\mathbf{X}})$ contains $\hat{X}_2$. Adding the two, $h(\mathbf{X} \mid \hat{\mathbf{X}}) \le h(X_1 \mid \hat{X}_1) + h(X_2 \mid \hat{X}_2)$. Combining the two terms,

$$I(\mathbf{X};\hat{\mathbf{X}}) = h(\mathbf{X}) - h(\mathbf{X} \mid \hat{\mathbf{X}}) \ge \sum_{i=1}^{2} \big(h(X_i) - h(X_i \mid \hat{X}_i)\big) = \sum_{i=1}^{2} I(X_i;\hat{X}_i). \qquad \blacksquare$$

(b) **(Converse.)** Let $p(\hat{\mathbf{x}} \mid \mathbf{x})$ be *any* test channel feasible for $R(D)$, and define $D_i := \mathbb{E}[(X_i - \hat{X}_i)^2]$ for the distortion it induces on each component. Summing, $D_1 + D_2 = \mathbb{E}\|\mathbf{X} - \hat{\mathbf{X}}\|^2 \le D$, so $(D_1, D_2)$ is a feasible split. The channel's marginal $X_i \to \hat{X}_i$ is itself a single-letter test channel achieving distortion $D_i$, and the scalar rate–distortion function $R_i(D_i)$ is by definition the *smallest* mutual information over all such channels; therefore $I(X_i;\hat{X}_i) \ge R_i(D_i)$. Chaining this with part (a),

$$I(\mathbf{X};\hat{\mathbf{X}}) \overset{(a)}{\ge} \sum_i I(X_i;\hat{X}_i) \;\ge\; \sum_i R_i(D_i) \;\ge\; \min_{D_1+D_2\le D} \sum_i R_i(D_i).$$

The last quantity is a constant independent of the chosen channel. Taking the minimum of the left-hand side over all feasible channels — which by definition is $R(D)$ — preserves the inequality:

$$R(D) \ge \min_{D_1+D_2\le D} \big(R_1(D_1) + R_2(D_2)\big). \qquad \blacksquare$$

(c) **(Achievability.)** Fix any split $(D_1, D_2)$ with $D_1 + D_2 \le D$; we may take $0 < D_i \le \sigma_i^2$, since raising $D_i$ above $\sigma_i^2$ leaves $R_i = 0$ while only loosening the distortion budget (and $D_i \to 0$ is the lossless limit, $R_i \to \infty$), so it never helps the minimum. For each component let $p_i^\star(\hat{x}_i \mid x_i)$ be the optimal scalar Gaussian test channel attaining $I(X_i;\hat{X}_i) = R_i(D_i)$ at distortion exactly $D_i$, and form the *product* channel

$$p(\hat{\mathbf{x}} \mid \mathbf{x}) = p_1^\star(\hat{x}_1 \mid x_1)\, p_2^\star(\hat{x}_2 \mid x_2).$$

Because both the source ($X_1 \perp X_2$) and the channel factorize, the two pairs $(X_1, \hat{X}_1)$ and $(X_2, \hat{X}_2)$ are mutually independent, so the joint law satisfies

$p(\mathbf{x}, \hat{\mathbf{x}}) = \prod_i p(x_i, \hat{x}_i)$. Mutual information of independent pairs is additive,

$$\begin{aligned} I(\mathbf{X}; \hat{\mathbf{X}}) &= \mathbb{E} \log \frac{p(\mathbf{X}, \hat{\mathbf{X}})}{p(\mathbf{X})\, p(\hat{\mathbf{X}})} = \sum_i \mathbb{E} \log \frac{p(X_i, \hat{X}_i)}{p(X_i)\, p(\hat{X}_i)} \\ &= \sum_i I(X_i; \hat{X}_i) = \sum_i R_i(D_i), \end{aligned}$$

and the distortion adds by linearity of expectation, $\mathbb{E}\|\mathbf{X} - \hat{\mathbf{X}}\|^2 = \sum_i \mathbb{E}[(X_i - \hat{X}_i)^2] = D_1 + D_2 \le D$. This product channel is therefore feasible for $R(D)$, so $R(D) \le \sum_i R_i(D_i)$. Since this holds for *every* feasible split, we may minimize the right-hand side:

$$R(D) \le \min_{D_1 + D_2 \le D} \sum_i R_i(D_i).$$

Together with the converse in part (b), the two bounds coincide:

$$R(D) = \min_{D_1 + D_2 \le D} \big(R_1(D_1) + R_2(D_2)\big). \qquad \blacksquare$$

Solving this convex program (Lagrange multipliers) yields the *reverse water-filling* allocation $D_i = \min(\theta, \sigma_i^2)$ — but that step is not required here.

3. **The $\chi^2$-Divergence: From Estimator Variance to the Evidence** (*35 points*) The $\chi^2$-divergence is $\chi^2(P\|Q) = \int \frac{(p-q)^2}{q} = \int \frac{p^2}{q} - 1$. After establishing its basic properties, you will use it twice: once to lower-bound the variance of an estimator, and once to upper-bound an intractable log-evidence.

   (a) **Foundations.** (*10 points*) Writing the likelihood ratio as $p/q$, show $\chi^2(P\|Q) = \mathrm{Var}_Q\big(\frac{p}{q}\big)$, and that $D(P\|Q) \le \chi^2(P\|Q)$.
   *Hint.* $\log t \le t - 1$.

   (b) **Hammersley–Chapman–Robbins.** (*15 points*) For any statistic $T$ with $\mathbb{E}_P[T] = a$, $\mathbb{E}_Q[T] = b$, prove

   $$\mathrm{Var}_Q(T) \;\ge\; \frac{(a-b)^2}{\chi^2(P\|Q)}.$$

   Then, for a parametric family $\{P_\theta\}$ with an unbiased estimator of $\theta$, take $Q = P_\theta$, $P = P_{\theta+h}$ and let $h \to 0$ to recover the Cramér–Rao bound $\mathrm{Var}_\theta(T) \ge 1/I(\theta)$.
   *Hint.* Express $a - b$ as a covariance under $Q$ and apply Cauchy–Schwarz.

   (c) **A $\chi^2$ upper bound on the evidence (CUBO).** (*10 points*) For a latent-variable model $p(x) = \int p(x,z)\, dz$, a proposal $q(z \mid x)$, and weight $w = \frac{p(x,z)}{q(z|x)}$, define $\mathrm{CUBO}_2 = \frac{1}{2} \log \mathbb{E}_{q(z|x)}[w^2]$. Show $\log p(x) \le \mathrm{CUBO}_2$, that the gap is exactly $\frac{1}{2} \log\big(1 + \chi^2(p(z \mid x) \| q(z \mid x))\big)$, and hence $\mathrm{ELBO} \le \log p(x) \le \mathrm{CUBO}_2$.
   *Hint.* $p(x) = \mathbb{E}_q[w]$ and $(\mathbb{E}_q[w])^2 \le \mathbb{E}_q[w^2]$; then write $w = p(x)\, \frac{p(z|x)}{q(z|x)}$.

**Solution: The $\chi^2$-Divergence.**

(a) **(Foundations.)** Let $W = p/q$ be the likelihood ratio. Its first two $Q$-moments are

$$\mathbb{E}_Q[W] = \int q\,\frac{p}{q} = \int p = 1, \qquad \mathbb{E}_Q[W^2] = \int q\left(\frac{p}{q}\right)^2 = \int \frac{p^2}{q}.$$

Hence $\mathrm{Var}_Q(W) = \mathbb{E}_Q[W^2] - (\mathbb{E}_Q[W])^2 = \int \frac{p^2}{q} - 1 = \chi^2(P\|Q)$, the first claim. For the comparison with KL, apply $\log t \le t - 1$ with $t = p/q$ pointwise:

$$D(P\|Q) = \int p \log\frac{p}{q} \;\le\; \int p\Big(\frac{p}{q} - 1\Big) = \int \frac{p^2}{q} - \int p = \int \frac{p^2}{q} - 1 = \chi^2(P\|Q).$$

(b) **(Hammersley–Chapman–Robbins.)** Assume $P \ne Q$ and $\chi^2(P\|Q) < \infty$, so the bound is non-vacuous. Write the mean gap as a $Q$-expectation against $W$. Since $\mathbb{E}_Q[W] = 1$,

$$\begin{aligned} a - b = \mathbb{E}_P[T] - \mathbb{E}_Q[T] &= \int T\,p - \int T\,q = \int T(p-q) \\ &= \int q\,T\Big(\frac{p}{q} - 1\Big) = \mathbb{E}_Q\big[T(W-1)\big]. \end{aligned}$$

Because $\mathbb{E}_Q[W-1] = 0$, this expectation is exactly a covariance: $a - b = \mathbb{E}_Q[T(W-1)] = \mathrm{Cov}_Q(T, W)$. By the Cauchy–Schwarz inequality for covariance,

$$(a-b)^2 = \mathrm{Cov}_Q(T,W)^2 \le \mathrm{Var}_Q(T)\,\mathrm{Var}_Q(W) = \mathrm{Var}_Q(T)\,\chi^2(P\|Q),$$

and rearranging gives $\mathrm{Var}_Q(T) \ge (a-b)^2/\chi^2(P\|Q)$.

*Cramér–Rao limit.* Assume $\{P_\theta\}$ is smooth in $\theta$ with finite Fisher information. Put $Q = P_\theta$, $P = P_{\theta+h}$. Unbiasedness means $\mathbb{E}_{\theta'}[T] = \theta'$, so $a = \theta + h$, $b = \theta$, and $a - b = h$; thus $\mathrm{Var}_\theta(T) \ge h^2/\chi^2(P_{\theta+h}\|P_\theta)$. Taylor-expanding the density, $p_{\theta+h} = p_\theta + h\,\partial_\theta p_\theta + O(h^2)$, so $(p_{\theta+h} - p_\theta)^2 = h^2(\partial_\theta p_\theta)^2 + O(h^3)$ and

$$\chi^2(P_{\theta+h}\|P_\theta) = \int \frac{(p_{\theta+h} - p_\theta)^2}{p_\theta} = h^2 \int \frac{(\partial_\theta p_\theta)^2}{p_\theta} + O(h^3) = h^2 I(\theta) + O(h^3),$$

where $I(\theta) = \int (\partial_\theta \log p_\theta)^2\, p_\theta$ is the Fisher information. Letting $h \to 0$,

$$\mathrm{Var}_\theta(T) \ge \frac{h^2}{h^2 I(\theta) + O(h^3)} \longrightarrow \frac{1}{I(\theta)}.$$

So HCR is a finite-difference Cramér–Rao bound, with $\chi^2$ in the role of (squared) Fisher distance. ■

(c) **(CUBO.)** Assume $q(z \mid x) > 0$ wherever $p(z \mid x) > 0$, and $\mathbb{E}_q[w^2] < \infty$ so that $\mathrm{CUBO}_2$ is finite. First, $p(x)$ is the $q$-average of the weight:

$$p(x) = \int p(x,z)\,dz = \int q(z \mid x)\,\frac{p(x,z)}{q(z \mid x)}\,dz = \mathbb{E}_q[w].$$

Since the square is convex, Jensen gives $p(x)^2 = (\mathbb{E}_q[w])^2 \le \mathbb{E}_q[w^2]$; taking $\frac{1}{2}\log$ (increasing) of both sides yields $\log p(x) \le \frac{1}{2}\log \mathbb{E}_q[w^2] = \mathrm{CUBO}_2$.

*Exact gap.* Factor the weight through the true posterior, $w = \frac{p(x,z)}{q(z|x)} = p(x)\,\frac{p(z|x)}{q(z|x)}$ (using $p(x,z) = p(x)\,p(z \mid x)$). Then

$$\mathbb{E}_q[w^2] = p(x)^2\,\mathbb{E}_q\Big[\Big(\frac{p(z \mid x)}{q(z \mid x)}\Big)^2\Big] = p(x)^2 \int q(z \mid x)\,\frac{p(z \mid x)^2}{q(z \mid x)^2}\,dz$$
$$= p(x)^2 \int \frac{p(z \mid x)^2}{q(z \mid x)}\,dz.$$

By the foundation identity $\int \frac{p^2}{q} = 1 + \chi^2(p\|q)$ from part (a), the last integral is $1 + \chi^2(p(z \mid x)\|q(z \mid x))$, so

$$\mathrm{CUBO}_2 = \tfrac{1}{2}\log \mathbb{E}_q[w^2] = \log p(x) + \tfrac{1}{2}\log\big(1 + \chi^2(p(z \mid x)\|q(z \mid x))\big).$$

The gap $\frac{1}{2}\log(1+\chi^2) \geq 0$, and vanishes iff $\chi^2 = 0$, i.e. $q(z \mid x) = p(z \mid x)$. Finally recall the ELBO is $\mathbb{E}_q[\log w] \leq \log \mathbb{E}_q[w] = \log p(x)$ (Jensen, concavity of log). Combining the two,

$$\mathrm{ELBO} \leq \log p(x) \leq \mathrm{CUBO}_2,$$

bracketing the intractable log-evidence between two computable bounds. The ELBO gap is the *reverse* KL $D(q(z \mid x)\|p(z \mid x))$ (mode-seeking), while the CUBO gap is driven by $\chi^2(p(z \mid x)\|q(z \mid x))$ (mass-covering). ■